

TractStar

Engineering and scientific handoff

A continuation brief for the merged prototype, first and second wave closure, and Wave 3 scientific model work.

Merged baseline: main at 304ee87 (PR #21)

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Repository: nathanwjclark/singing-teacher

This document distinguishes runnable software from generated evidence, physical-device evidence and scientific validation. An inaccurate or non-identifiable result is acceptable; an unsupported anatomical claim is not.

The product objective

TractStar is an open-source singing research prototype. Its distinctive goal is to let a singer provide voice recordings, images/video and optional depth or acoustic-probe evidence; maintain competing hypotheses about that singer's vocal acoustics and visible anatomy; ask Astra to choose a safe, supported experiment; record the attempt and subjective sensation; compare the result with a prediction frozen before the attempt; and update or retain hypotheses with an explicit reason.

The product is a research harness for a moonshot. It may produce an inaccurate or non-identifiable model. A faithful discrepancy, a retained ambiguity and a useful next experiment are valid outcomes. The system must never turn a native template surface, a camera landmark, a microphone feature, a simulator control or an educational illustration into a claim that hidden anatomy, vocal-fold contact, muscle activation or clinical function was measured.

The three-wave framing is:

- 1 **Prototype foundation:** capture and provenance, native vocal-tract forward modeling, finite PCM fitting, session ledger, browser app and graceful degradation.
- 2 **Connected research loop:** Astra decisions, frozen prospective experiments, outcomes and model lineage, optional source/phonation analysis, motion/audio comparisons, rear LiDAR conditional fusion, visual likelihood, illustrated teaching, probe import/fit, replay/export and visible tongue tooling.
- 3 **Scientific improvement:** richer spectral objectives, cue-to-execution learning, time-resolved articulation, self-oscillating source mechanics, coupled oral/nasal probe physics, reproducible scoring, phrase-level transfer and controlled modality/accuracy evaluations.

What is actually merged on main

PR #19 merged the rear-LiDAR conditional likelihood and competing source bank. Original depth is projected through calibrated rays to explicitly annotated outer-lip pixels; retained hypotheses are ranked and the authoritative session can adopt a new model. The UI previews only hash-verified adopted native geometry. Rejected, stale, failed and duplicate attempts remain visible. Source alternatives are frozen before later recordings, scored against one canonical original frame and retained in a separate conditional ranking. Recovery includes concurrent-request exclusion, terminal geometry-export retry identities and stale unsubmitted-intent release.

PR #20 merged conditional visual likelihood and illustrated physiology teaching. Original video frames can be annotated in pixels with a fixed declared camera candidate; later frames are scored as visible, occluded or missing. Results reach bounded Astra context and session export without anatomical adoption. Teaching has separate general-explanation, model-prediction and recorded-result modes, reviewed mechanisms, static/reduced-motion fallbacks, native geometry/audio comparisons and explicit recorded-outcome handling. The supported personalized comparison is the committed tongue/jaw vowel operator. Cricothyroid, soft-palate and other unsupported mechanisms remain educational-only.

PR #21 added final native teaching/browser acceptance and a guard for incomplete historical scientific receipts. The real acceptance fixture exercised native geometry, explicit A/B synthesis playback, a recorded numerical outcome, historical and educational views, reduced motion, mobile layout and complete app mounting in 18.9 seconds. The fixture used generated evidence and a seeded software Astra decision; it made no paid provider call.

The merged prototype also contains:

- canonical microphone measurements and an optional live phonation panel;
- an isolated source/tract research fitter and bounded competing source bank;
- native iPhone capture/import contracts and a native iOS acquisition app that is unsigned here;

- motion recording, audio-only conditional analysis and bounded temporal comparisons;
- acoustic-probe import and oral external-drive fitting with graceful missing-calibration behavior;
- personal sensation memory, cue/recall/transfer records and session export;
- a local server with server-side Astra credentials, finite call budgets and localhost-only experimental routes;
- original-byte/hash verification, immutable job/session lineage and restart/retry handling.

What the merged prototype does not establish

No merged code proves complete internal anatomy reconstruction, calibrated posterior probabilities, vocal-fold closure/contact, tissue mechanics, nasal-cavity geometry, cricothyroid motion, hidden tongue shape or coaching efficacy. The optional phonation path measures source-plus-filter acoustics and can rank prescribed source alternatives; it does not diagnose closure. The LiDAR path constrains an outer-lip distance likelihood; it does not see the palate, larynx, vocal folds or internal muscles. The visual path is a conditional pixel comparison with declared correspondences and camera assumptions. All generated/native fixtures are software evidence, not human evidence.

Remaining prototype and first/second-wave closure

The core prototype foundation is runnable on `main`, but the connected research loop still has a few integration and acceptance items before it should be called complete:

- **App-managed probe setup:** finish the in-app calibration-package flow and its browser/native checks. A metadata package must never stand in for a calibrated loudspeaker, microphone, placement or timing measurement.
- **Replay/recompute:** register the read-only recompute route and export checks, then verify that a retained recording can be recomputed without changing the historical forecast or inventing absent media.
- **Melodic transfer:** finish the bounded 2-8-note phrase target and verify missing-voicing, timing and unsupported-phrase behavior. Single-note scoring must remain backwards compatible.
- **Visible tongue baseline:** review and merge the public region detector only as visible-region evidence. Keep the private dynamic diagnostic and any hidden- tongue interpretation out of the physiological model.
- **Physical/device acceptance:** run signed iPhone 13 Pro Max and iPhone 15 Pro capture, rear-LiDAR calibration, real probe playback and at least two real Astra-guided loops. The current host lacks the full Xcode/iOS SDK, so this is a separate device lane rather than a local build assumption.

These are software, device and evidence gates owned by the next integration agent; they do not require claiming that the scientific inverse problem is solved. A lane is complete only when its contract, exact command, artifacts, test result, evidence source and limitations are recorded in the handoff and current-status document.

Current unmerged Wave 3 worktrees

These are research branches, not part of `main`. Preserve their commits and dirty edits; do not cherry-pick blindly across shared files.

Worktree / branch	Known checkpoint	State and next action
singing-teacher-wave3-integration / codex/wave3-integration	f0868ba	Root integration scratch branch. Includes merged tongue work and bounded motion-context summarization plus explicit objective/session hooks. Review, then use as the integration base for selected Wave 3 PRs.

Worktree / branch	Known checkpoint	State and next action
singing-teacher-wave3-spectral / codex/wave3-spectral	8c68f60, 5813a67 plus dirty edits	Multi-resolution log-spectrum objective, prospective policy pinning and native/app integration are partly implemented. Finish dirty <code>live_capture_jobs.py</code> / <code>pcm_design.py</code> , verify exact frame binding and run independent evaluation.
singing-teacher-wave3-eval / codex/wave3-eval	57907c0	Frozen independent evaluation protocol and first report. The unmodified result is mixed/inconclusive: gain perturbation improves robustness, unseen-vowel generalization is poor, and noise/out-of-support cases can be worse. Do not retune on these held-out cases.
singing-teacher-wave3-control / codex/wave3-control	304ee87 plus untracked <code>control_pcm.py</code>	Architecture handoff is complete. Implement exact cue/context bindings, finite PCM execution banks, repeated-attempt support and residual calibration; root must add shared session/service/Astra/export hooks.
singing-teacher-wave3-tongue / codex/wave3-tongue	dc0cf51	Public TongueSAM region baseline, separate region metrics/exports, license/hash verification and browser QA passed. It was merged into the integration scratch branch but not main; submit as a focused PR after review. It remains visible-region evidence only.
singing-teacher-wave3-source / codex/wave3-source	304ee87 plus dirty edits	Native two-mass source-family investigation is underway. It needs a guarded engine family switch, provenance restoration and equal-budget held-out comparison before integration.
singing-teacher-wave3-motion / codex/wave3-motion	7bd8c58 plus <code>.local-tests</code>	Time-resolved conditional audio bank and uncertainty sets are implemented and native generated tests passed. Finish HTTP/browser verification, clean test artifacts and review source/articulation confounding.
singing-teacher-wave3-probe / codex/wave3-probe	304ee87 plus dirty edits	In-app calibration-package setup is implemented in progress. Finish tests/docs, then integrate without allowing metadata to substitute for calibration evidence.
singing-teacher-wave3-replay / codex/wave3-replay	37d4d54, 77798af plus dirty tests	Read-only batch recompute route/UI and spectral policy work exist. Register route/export checks on the integration branch and finish actual HTTP/browser verification.
singing-teacher-wave3-phrase / codex/wave3-phrase	304ee87 plus dirty edits	Short 2-8-note frozen melody scoring is in progress. Finish browser QA, then review timing/voicing/missing-data semantics.

The old worktree inventory also contains many historical agent branches. They are not a task board. Use `git log`, the relevant plan and this handoff before reviving any one.

Wave 3 scientific work already learned

The native library exposes a real two-mass glottis family in addition to the geometric glottis. A guarded family switch can produce distinct dynamic waveforms, but requested F0 and observed F0 may differ; that difference must be recorded. It is not evidence of measured fold mechanics.

The native library also exposes full tract sections and transmission-line matrices covering pharynx, mouth, nose, fossa and sinuses. A five-equation oral/nasal external-drive solve has passed numerical reciprocity and residual checks, including exact zero nasal flow for a rigid sealed boundary. The next implementation should bind those native matrices and solve the coupled network using real native geometry, explicit CGS->SI conversion, frequency-grid bounds and declared source/receiver coupling. Do not replace the native nasal branch with a guessed tube or equate velum opening with nostril occlusion.

The first independent spectral evaluation is deliberately mixed: the richer objective is more robust to gain perturbation, but unseen-vowel generalization is weak, noise/out-of-support cases can degrade, and some predictions are unavailable or tied. This is exactly the evidence boundary the next agent must preserve. Wave 3 is not a license to call the new objective

"more accurate" without a fresh held-out comparison.

Prioritized continuation plan

- 1 Integrate the spectral branch only after its policy, exact original-frame binding, legacy compatibility and independent report are complete. Compare coarse and multi-resolution objectives at equal native budgets; retain both scores.
- 2 Implement cue-to-execution learning. Freeze the exact delivered cue and context before each attempt, match history by full anatomy/native/extractor/profile/cue/context identity rather than model ID, score a finite PCM control bank, and use at least three matched attempts to weight the next frozen forecast. Keep empirical microphone residuals separate from inferred physical controls.
- 3 Complete the two-mass source-family adapter and evaluate source-family selection on unseen pitch/vowel/source conditions. Do not call prescribed source parameters vocal-fold closure or muscle activity.
- 4 Integrate the time-resolved motion bank and the coupled oral/nasal probe operator. Preserve unvoiced, noisy, occluded and out-of-support windows as explicit missing evidence.
- 5 Complete in-app probe setup and read-only replay/recompute, then add the melodic transfer protocol. Each must have actual browser/native acceptance and a clear unsupported path.
- 6 Merge the tongue baseline as a separate visible-region feature after license, asset-hash and model-fallback review.
- 7 Run physical acceptance on both available iPhones: signed iOS build, permissions/interruption/camera-return, real depth/RGB/PCM timing, known-target calibration, actual acoustic-probe calibration and two real Astra-guided loops. The current host has Command Line Tools but no full Xcode/iOS SDK (xcodebuild unavailable).
- 8 Run the scientific evaluation matrix: held-out humans/reference objects, anatomy/source/articulation compensation, room/device perturbations, audio vs audio+RGB vs audio+RGB+depth at equal budgets, model mismatch, identifiability and cue-free phrase transfer. Positive recovery is not required; honest negative/inconclusive results are required.

Shared integration contract

Every handoff must include: commit, branch, exact command, contract/version, expected artifacts, actual checks, evidence source (synthetic, reference object or human), scientific outcome (untested, positive, negative or inconclusive) and known limitations. New forecasts are frozen before target recordings. Model updates and no-change decisions retain parent IDs and artifact hashes. Legacy receipts must remain explicitly unverified when their scorer policy was not pinned.

The next agent should start from a clean main, create a new worktree per feature, integrate one lane at a time, run build/lint plus targeted tests, then run the real browser/native acceptance before merging. Keep API keys in the private .env; no handoff document should contain them. Never treat a generated fixture or a successful plumbing test as human anatomical validation.

Useful commands

```
cd "/Users/nicolelu/Documents/ChatGPT/New project/singing-teacher-repo"
git fetch origin
git switch main
git pull --ff-only origin main
npm install
npm run build
npm run lint
npm test
OPENAI_ENV_FILE="/path/to/private/.env" \
LOCAL_DATA_DIR="/path/to/private/.local-data" \
SCIENCE_DATA_DIR="/path/to/private/.local-data/science-jobs" \
```

```
PHONATION_SOURCE_ENABLED=1 PHONATION_MEASUREMENT_ENABLED=1 \  
LIDAR_PREVIEW_ENABLED=1 LIDAR_FUSION_ENABLED=1 \  
VISUAL_LIKELIHOOD_ENABLED=1 VISUAL_TEACHING_ENABLED=1 \  
VISUAL_TEACHING_MODEL_ENABLED=1 VISUAL_TEACHING_AUDIO_ENABLED=1 \  
npm run science:local
```

For Wave 3 Python checks, use the existing shared environment when available:

```
PYTHONPATH=./science/src science/.venv/bin/python -m pytest <targeted-tests> -q
```

The OPENAI_ENV_FILE value above is a path placeholder. Never commit the actual key, copy it into a report, or run broad live-provider loops while developing numerical code.

Documentation map

- Current merged implementation status
- Delivery and recovery audit
- Wave 3 scientific plan
- Research-harness completion definition
- Astra execution
- LiDAR status
- Phonation/source status
- Visual/teaching status
- Probe app contract
- Replay limitations

Historical plans and reviews remain useful for rationale, but this handoff and the current-status document are authoritative for what is merged today.